

Inference for Weak OU Mean Reversion: A Reproducible Finite-Horizon Study

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Abstract

A short trajectory can provide little information about weak mean reversion, even when observations are frequent. This study examines that difficulty for a scalar Ornstein–Uhlenbeck process with unknown drift and diffusion scale, conditional on a fixed initial state. It derives the exact transition likelihood, profiles the nuisance parameters analytically, and implements a chronological split-profile confidence rule based on universal inference. A restricted continuous-path Kullback–Leibler calculation distinguishes an information limit from the behavior of a particular procedure. In a saved development experiment with 1,600 trajectories, the split rule has true-parameter membership between 0.99 and 1.00 but excludes zero in at most 1% of each cell. An uncalibrated chi-square likelihood-ratio diagnostic has membership between 0.57 and 0.91. The Julia implementation is checked against frozen Python and separately coded base-R references. The result is a reproducible technical and numerical study; it does not introduce a new inference method or establish certified confidence endpoints or a calibrated bootstrap comparison.

1 Introduction

For a persistent continuous-time process, a positive fitted mean-reversion coefficient does not by itself demonstrate that reversion is well identified. The relevant inferential questions include whether zero can be excluded, whether arbitrarily slow reversion remains plausible, and whether uncertainty has a finite upper endpoint. These questions become especially delicate when both the drift intercept and diffusion scale must be estimated from one finite observation window.

This report studies

$$dX_t = (a - \kappa X_t) dt + \sigma dW_t, \quad a \in \mathbb{R}, \quad \kappa \geq 0, \quad \sigma > 0, \quad (1)$$

conditional on an observed, fixed $X_0 = x_0$. Observation times are deterministic, or exogenous so that conditioning on the schedule preserves the stated transition law. At $\kappa = 0$ the model is Brownian motion with drift a . For $\kappa > 0$, the long-run mean is $\theta = a/\kappa$; it is not identified at zero. A bounded- θ restriction would force $a \rightarrow 0$ as $\kappa \rightarrow 0$ and would therefore change the boundary experiment.

Diffusion drift bias and persistent-process inference have substantial prior literature [1, 2]. Hansen’s grid bootstrap and its continuous-time extension by Lui, Xiao and Yu supply important comparison methods [3, 4]. The split-profile construction below is a specialization of universal inference [5]. The present objective is to verify a boundary-aware implementation and expose limitations that a subsequent calibrated comparison must address. Migration to Julia is an implementation choice, not a methodological contribution.

2 Exact likelihood and inferential methods

2.1 Transition law and nuisance profiling

Let $0 = t_0 < t_1 < \dots < t_n = T$ and $\Delta_i = t_i - t_{i-1} > 0$. Solving the linear stochastic differential equation gives

$$X_i \mid X_{i-1} \sim N\left(\phi_i X_{i-1} + ab_i, \sigma^2 v_i\right), \quad \phi_i = e^{-\kappa \Delta_i}, \quad b_i = \frac{1 - e^{-\kappa \Delta_i}}{\kappa}, \quad v_i = \frac{1 - e^{-2\kappa \Delta_i}}{2\kappa}. \quad (2)$$

At zero, the continuous limits are exactly $\phi_i = 1$ and $b_i = v_i = \Delta_i$. The implementation uses stable exponential expressions and an explicit zero branch, with no Euler approximation or positive lower bound on κ . There is no stationary initial density.

For a candidate κ , write $r_i = x_i - \phi_i x_{i-1}$ and $s^2 = \sigma^2$. The conditional log likelihood is

$$\ell(\kappa, a, s^2) = -\frac{1}{2} \sum_{i=1}^n \left\{ \log(2\pi s^2 v_i) + \frac{(r_i - ab_i)^2}{s^2 v_i} \right\}.$$

Weighted least squares and one-dimensional variance maximization yield

$$\hat{a}_\kappa = \frac{\sum_i b_i r_i / v_i}{\sum_i b_i^2 / v_i}, \quad R_\kappa = \sum_i \frac{(r_i - \hat{a}_\kappa b_i)^2}{v_i}, \quad (3)$$

$$\hat{\sigma}_\kappa^2 = \frac{R_\kappa}{n}, \quad \ell_p(\kappa) = -\frac{1}{2} \left[n \{ \log(2\pi \hat{\sigma}_\kappa^2) + 1 \} + \sum_i \log v_i \right]. \quad (4)$$

Indeed, the residual quadratic is $R_\kappa + (a - \hat{a}_\kappa)^2 \sum_i b_i^2 / v_i$, with a strictly positive coefficient. When $R_\kappa > 0$, differentiating the remaining likelihood in s^2 gives the maximum at R_κ / n . The divisor counts transitions; it is not the residual degrees-of-freedom divisor of an unbiased variance estimator. When $R_\kappa = 0$, the likelihood supremum is infinite as $s^2 \downarrow 0$. This degeneracy is retained rather than replaced by a variance floor.

The numerical fit evaluates zero and a positive logarithmic grid up to $200/T$, then refines local candidates by a bounded scalar search. This is not a certified global maximum. As $\kappa \rightarrow \infty$, the profiled likelihood tends to the free-mean, free-variance iid Gaussian likelihood of $x_1, \dots, x_n$. To see this, reparameterize $a = \kappa\beta$ and $\sigma^2 = 2\kappa\tau^2$ in (2); the means tend to β and variances to τ^2 . The explicit weighted least-squares profiles converge to the successor sample mean and variance, giving convergence of the profiled likelihood as well. If the observed successor states are all equal, the limiting supremum is infinite. Cap contact and comparison with this tail limit are reported as warnings. A search cap is never interpreted as a confidence endpoint.

2.2 An explicitly uncalibrated likelihood-ratio diagnostic

Let $\tilde{\ell}_{\max}$ denote the best value returned by the bounded search. The diagnostic statistic is

$$D(\kappa) = 2\{\tilde{\ell}_{\max} - \ell_p(\kappa)\}.$$

The development comparison accepts a candidate when $D(\kappa) \leq \chi_{1,0.95}^2 = 3.84145882$. This cutoff is retained to examine its behavior; it is not justified here by Wilks' theorem. Neither an ordinary chi-square limit nor a simple boundary mixture is assumed reliable in the fixed-span, weak-reversion regime. Approximate global optimization is an additional numerical limitation of this diagnostic.

2.3 Chronological split-profile rule

Choose a fixed split after m transitions. Fit an OU density on the prefix $X_0, \dots, X_m$, freeze its parameters, and evaluate the remaining transition product $q(X_{m+1:n} \mid \mathcal{F}_m)$, where $\mathcal{F}_m = \sigma(X_0, \dots, X_m)$ with the schedule conditioned on. For each realized prefix, q must be a normalized conditional density. Define

$$E_\kappa = \frac{q(X_{m+1:n} \mid \mathcal{F}_m)}{\sup_{a \in \mathbb{R}, \sigma > 0} p_{\kappa, a, \sigma}(X_{m+1:n} \mid X_m)}, \quad C_\alpha = \{\kappa \geq 0 : E_\kappa \leq 1/\alpha\}. \quad (5)$$

The denominator is the analytic nuisance supremum on the suffix. At the true parameters $(\kappa_0, a_0, \sigma_0)$, conditional-density domination gives

$$\mathbb{E}_{\kappa_0, a_0, \sigma_0}[E_{\kappa_0} \mid \mathcal{F}_m] \leq \int q(y \mid \mathcal{F}_m) dy = 1.$$

Markov's inequality therefore implies $P_{\kappa_0, a_0, \sigma_0}(\kappa_0 \in C_\alpha) \geq 1 - \alpha$. Conditioning on X_m handles prefix/suffix dependence; independence of the two segments is not required. Approximate prefix fitting preserves this argument when it still produces a normalized positive-variance density. An underestimated denominator, future-informed fitting, or an unreported training failure does not inherit the guarantee.

This is the conditional/profile-likelihood principle of universal inference [5]. Its mathematical guarantee concerns the exact model and exact nuisance supremum. Floating-point agreement is not a certificate that every computed denominator is a conservative upper bound. The rule is fixed-horizon and model-based, not a confidence sequence, prediction interval, or distribution-free guarantee.

3 Information limitations over a finite horizon

3.1 Dimensionless parameters

Brownian scaling of $U_s = (X_{Ts} - x_0)/(\sigma\sqrt{T})$ gives

$$dU_s = (\delta - cU_s) ds + dB_s, \quad c = \kappa T, \quad \delta = \frac{(a - \kappa x_0)\sqrt{T}}{\sigma}.$$

Thus c measures reversion relative to the observation span, but does not by itself determine every experiment: δ and the normalized schedule also matter. Increasing n at fixed T is an infill experiment; increasing T at a fixed observation step is a different information change.

3.2 A restricted pathwise KL calculation

For this subsection only, assume $a = 0$, $x_0 = 0$, and the same known σ under both laws. Let P_κ and P_0 denote continuous-path laws on $[0, T]$ for OU and driftless Brownian motion. Under P_0 , $X_t = \sigma W_t$. Itô's formula expresses the candidate likelihood ratio as

$$L_t = \exp \left\{ -\frac{\kappa}{\sigma} \int_0^t X_s dW_s - \frac{\kappa^2}{2\sigma^2} \int_0^t X_s^2 ds \right\} = \exp \left\{ \frac{\kappa t}{2} - \frac{\kappa X_t^2}{2\sigma^2} - \frac{\kappa^2}{2\sigma^2} \int_0^t X_s^2 ds \right\}.$$

For $\kappa \geq 0$, this positive local martingale is bounded by $e^{\kappa T/2}$ on the finite horizon. Localization and dominated convergence make it a true martingale. Girsanov's theorem then identifies $L_T = dP_\kappa/dP_0$. Under P_κ , write

$$\log L_T = -\frac{\kappa}{\sigma} \int_0^T X_t dW_t^{(\kappa)} + \frac{\kappa^2}{2\sigma^2} \int_0^T X_t^2 dt,$$

where $W^{(\kappa)}$ is Brownian motion under P_κ . Its stochastic integral has finite second moment and hence mean zero under this measure. Using $\mathbb{E}_\kappa X_t^2 = \sigma^2(1 - e^{-2\kappa t})/(2\kappa)$ gives

$$\text{KL}(P_\kappa \| P_0) = \frac{\kappa^2}{2\sigma^2} \int_0^T \mathbb{E}_\kappa X_t^2 dt \quad (6)$$

$$= \frac{c}{4} - \frac{1 - e^{-2c}}{8} = \frac{c^2}{4} - \frac{c^3}{6} + O(c^4). \quad (7)$$

The identity is continuous at $c = 0$.

For discrete observations, the Gaussian chain rule gives

$$\text{KL}(P_\kappa^{\text{obs}} \| P_0^{\text{obs}}) = \frac{1}{2} \sum_i \left\{ \frac{v_i}{\Delta_i} - 1 - \log \frac{v_i}{\Delta_i} + \frac{(\phi_i - 1)^2 V_{i-1}}{\Delta_i} \right\}, \quad V_{i-1} = \frac{1 - e^{-2\kappa t_{i-1}}}{2\kappa}. \quad (8)$$

At $\kappa = 0$, take $V_{i-1} = t_{i-1}$ by continuity. Sampling is a measurable function of the full path. Data processing therefore bounds sampled KL by path KL and implies monotonicity for nested refinement. Any test of the restricted Brownian null with size at most α has power bounded by

$$\text{Power} \leq \min \left\{ 1, \alpha + \sqrt{\text{KL}(P_\kappa \| P_0)/2} \right\}. \quad (9)$$

This follows from total variation and Pinsker's inequality. At $c = 0.1$, path KL is approximately 0.0023413441, so the level-0.05 power bound is 0.0842151. Arbitrarily dense sampling cannot exceed the full-path information in this restricted experiment. The bound is loose and need not be attainable; it is not a power curve for the unknown-nuisance split rule. A difference between that rule's power and this envelope does not identify the cost of splitting.

Table 1: Restricted information calculation with $a = x_0 = 0$ and known σ . Discrete KL uses eight equal gaps; the final column is the continuous-path level-0.05 power upper bound. These are deterministic values, not simulations.

c	Sampled KL ($n = 8$)	Path KL	Power upper bound
0.1	0.002341315	0.002341344	0.084215
1.0	0.141745295	0.141916910	0.316380
4.0	0.860037053	0.875041933	0.711454

4 Development experimental design

The main empirical results use a saved Julia development pilot with $T = 1$, $a = x_0 = 0$, $\sigma = 1$, $\kappa \in \{0, 0.1, 1, 4\}$, and $n \in \{64, 256\}$ transitions. Regular gaps are $1/n$. The deterministic alternating schedule repeats $0.25/n$ and $1.75/n$, preserving the same T and n . This gives 16 cells, each with $R = 100$ independently simulated trajectories, for 1,600 trajectories in total. Both a and σ are

unknown to the inference procedures, even though their generating values are fixed. The split is $m = n/2$ and $\alpha = 0.05$.

Exact transition simulation uses a fresh Julia Xoshiro generator for each trajectory. The base seed is 2026092803; the stored seed equals this value plus the trajectory index in the declared loop order. Different resolutions and schedules use different paths. These comparisons are therefore not paired nested-path infill experiments. A future infill design must derive coarse observations from a common exact fine trajectory; longer-span comparisons should use common path prefixes.

For each trajectory, the pilot evaluates true-parameter membership and exclusion of zero directly from the two statistics. It does not construct full confidence sets or measure their widths. With I_r the relevant binary outcome, $\hat{p} = R^{-1} \sum_r I_r$ and the reported plug-in Monte Carlo standard error is $\sqrt{\hat{p}(1 - \hat{p})/R}$. The uncertainty unit is one independent trajectory, not an observation within a trajectory. Pointwise 95% Wilson intervals are also reported, including at zero or one observed rates; these describe simulation uncertainty, not parameter confidence sets for κ .

All attempted trajectories remain in denominators. If S observed successes and F unresolved trajectories occur in R attempts, the empirical success fraction is bounded by $[S/R, (S + F)/R]$, with failures reported separately. The saved Julia pilot has zero failed trajectories and no recorded cap, tail, optimizer or numerical warnings. This does not certify global optimization. The entire pilot is exploratory development evidence, with no locked confirmation design or method selection justified by these results.

5 Results and numerical verification

5.1 Coverage alone does not establish informativeness

Table 2 reports the regular $n = 256$ cells; Figure 1 shows true-parameter membership in all 16 cells. The chi-square LR diagnostic has membership between 0.57 and 0.91, whereas split-profile membership is between 0.99 and 1.00. At the four $\kappa = 0$ cells, the LR diagnostic falsely excludes zero in 28%, 43%, 30% and 34% of trajectories (regular $n = 64, 256$, then alternating $n = 64, 256$). This is strong evidence of poor calibration for that particular cutoff in the tested development settings; it does not show that all likelihood-based inference fails.

Table 2: Selected Julia development results: regular schedule, $n = 256$ and 100 trajectories per row. Membership is at the true κ ; zero exclusion is a false rejection at $\kappa = 0$ and empirical power otherwise. LR uses an uncalibrated cutoff. Fitted κ values use bounded numerical search.

κ	Mean $\hat{\kappa}$	Truth membership		Zero exclusion	
		LR	Split	LR	Split
0.0	6.898	0.57	1.00	0.43	0.00
0.1	5.307	0.65	1.00	0.36	0.00
1.0	6.371	0.84	0.99	0.35	0.01
4.0	8.159	0.91	1.00	0.46	0.00

The split rule excludes zero in at most one of the 100 trajectories in any cell, including at $c = 4$. No exclusions in a cell give the Wilson interval $[0, 0.03699]$; 100 memberships give $[0.96301, 1]$. Neither zero estimated MCSE at these extremes nor very high membership certifies exact empirical risk. The inability to exclude zero is a substantive negative informativeness result. It motivates

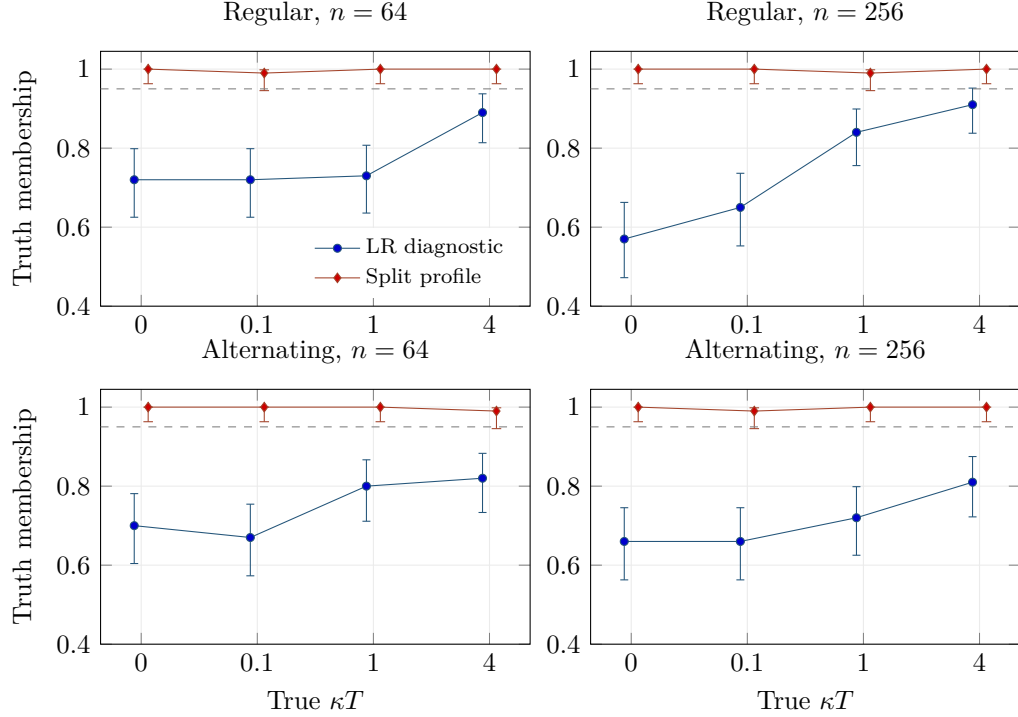


Figure 1: True-parameter membership in the Julia development pilot. Bars are pointwise 95% Wilson intervals from 100 independent trajectories per cell; the dashed line is 0.95. Horizontal positions are categorical, slightly offset for legibility; connecting segments guide the eye and do not interpolate power. Blue circles show the uncalibrated LR diagnostic; rust diamonds show split profile.

separating intrinsic weak information, nuisance estimation and chronological splitting, rather than treating near-perfect membership as success.

The earlier frozen Python development pilot independently showed the same qualitative pattern: LR membership 0.63–0.88, split membership 0.99–1, and split zero exclusion 0–0.01. Its seed family and language RNG differ; these are separate development runs, not paired trajectories or identical stochastic replications. Their rates are not pooled. The smaller 32-trajectory Julia smoke run validates the pipeline and is not the empirical study reported in Table 2.

5.2 Cross-language and numerical checks

The recorded Julia suite passes 2,468 assertions, including 2,236 deterministic scalar comparisons with frozen Python and independently coded base R. These counts are numerical checks, not independent Monte Carlo replications. The comparisons cover transition coefficients, conditional means and variances, likelihood, profiled nuisance estimates, selected LR statistics, and information calculations. Explicit innovations fix cross-language simulation paths. Fourteen further assertions in the suite use 256-bit calculations to check difficult small-KL cases with relative tolerance 10^{-12} and zero absolute tolerance.

The largest absolute discrepancy in closed-form comparisons is 3.553×10^{-14} . Bounded objective/LR comparisons differ by at most 3.242×10^{-14} ; bounded parameters and predictive/split

scores differ by at most 4.057×10^{-6} , within their predeclared 2×10^{-5} absolute and relative tolerances. Different scalar optimizers explain why a predictive score can be more sensitive than the optimized training likelihood. The ten independent R nuisance optimizations agree in objective to 6.359×10^{-13} . Tolerances were not enlarged after observing failures.

Independent checking found cancellation in the original sampled-KL computation: forming $u = v_i/\Delta_i - 1$ loses its first-order term for tiny $z = \kappa\Delta_i$. Julia instead evaluates

$$u = -z + \frac{2}{3}z^2 - \frac{1}{3}z^3 + \frac{2}{15}z^4 + O(z^5)$$

directly in the small-argument regime and evaluates $u - \log(1 + u)$ stably. For $\kappa = 10^{-16}$, $T = 1$ and one transition, the old expression gives about 3.0815×10^{-33} , exceeding the path KL of about 2.5×10^{-33} . The corrected calculation gives about 2.5×10^{-33} . The frozen references remain unchanged, and the discrepancy is documented. Even at the fixture value $\kappa = 10^{-10}$, the old cancellation causes about 5.1×10^{-7} relative error in a KL of order 10^{-20} ; absolute tolerance alone would conceal it. The separate high-precision checks address that issue. This repair is numerical, not a new statistical result.

Other tests cover Brownian limits, transition aggregation, affine state and time rescaling, explicit RNG reproducibility, prefix-only fitting, degeneracy and failure reporting. Simulator moment checks use 12,000 independent short paths in each of six cells, with six-standard-error thresholds. Those 72,000 paths check the simulator; they are not an additional inferential coverage study.

6 Limitations and next research step

The empirical design fixes zero drift, zero initial state, unit scale and unit horizon. It therefore does not establish performance over a nuisance grid, under longer spans, under endogenous sampling, or with observation noise. Different n values are not paired on nested paths. One hundred trajectories per cell permit only limited Monte Carlo precision, and the results are not locked confirmation evidence.

The study has no faithfully reproduced grid-bootstrap baseline. A comparison must first specify the paper version, null-imposed nuisance fitting, initialization, resampling law and inversion statistic [3, 4]. The known-long-run-mean and $\kappa(\mu - X_t)$ parameterizations in relevant literature must not be silently identified with unrestricted a at $\kappa = 0$. An ordinary plug-in bootstrap is not automatically finite-sample exact. The version-specific overlap and remaining full-text access gaps are recorded in the literature documentation; no novelty conclusion is inferred from an incomplete search.

Full inversion is also unfinished. The set in (5) need not be a single finite interval: zero, disconnected components, unbounded tails and numerical nonresolution must all remain possible. A finite accepted grid is not a certified continuum set. For any set C , half-life uncertainty is

$$H(C) = \{\log(2)/\kappa : \kappa \in C, \kappa > 0\} \cup \{+\infty : 0 \in C\}.$$

Thus accepting zero retains infinite half-life; a positive unbounded component approaches half-life zero without attaining it. Numerical tail and crossing certification is required before reporting finite endpoints or widths.

The next step is a faithful grid-bootstrap reproduction followed by confidence-set geometry, common-path infill and longer-span comparisons, and a sensitivity design in $(c, \delta, \text{schedule})$. The current derivations are elementary specializations of established theory; numerical agreement and

exploratory findings are separate evidence. No new method, efficiency theorem or superiority claim is established by this report.

7 Reproducibility

The repository provides Julia source, a locked project environment, numerical tests, deterministic reference fixtures and saved development outputs. The report’s tables and plots are generated from those outputs and embedded in this standalone L^AT_EX file. It compiles without external figures, data files or bibliography files. Julia is canonical; Python is a frozen numerical reference and base R supplies selected independent checks. The public tests use saved synthetic fixtures and require no Python installation.

Run record. Julia 1.12.5, one Julia thread and one BLAS thread; 100 independent trajectories in each of 16 cells; Xoshiro base seed 2026092803. The saved development run is

20260928T030948Z-julia-pilot-a8881b95.

Its recorded implementation and environment hashes match the initial source release d08dc35. The run itself preceded that commit and records its Git status as uncommitted; the match is established by the saved content hashes. The 32-path smoke, 1,600-path Python pilot and simulator moment checks have separate records and purposes.

The README gives installation and execution commands. To regenerate this report’s source from its preserved inputs, run

julia --startup-file=no --project=. scripts/build_report.jl

The report manifest records the template, builder, data and source identities. New runs create unique output directories; they do not overwrite archived results. The following SHA-256 fingerprints the declared data and implementation inputs, not elapsed-time reproducibility or the finished PDF:

ccebde7d4a4263a6830eb88875ff976acbaa3bb8237645ae1b91483873c748c1

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